

Submittal

Project Name:

10811 S ave C (Revision 1)

Contractor: Blacktech Solutions Corp

Engineer:

Architect:

Rep/Distributor:

Project Detail:

Customer: Tunisha Riley

Address: 10811 S Ave C

City: Chicago

State: IL

Zip: 60617

Submittal Date:

1/2/2025

Submitted By:

Name: Derrell Black

Company: Blacktech Solution corp

Email: Derrellblack@blacktechsolutionscorp.com

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Submittal Stage: For Approval

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Notes

Equipment Bill Of Material

Quantities

Qty	Model	Description	Tag
2	stock controller	Wireless remote controller	
1	MXZ-2C20NA4-U1	NewR410A MXZ Series Outdoor Unit	OutdoorUnit 1
2	MFZ-KJ09NA-U1	Floor-Standing Type (Exposed) Indoor Unit	IndoorUnit 1,IndoorUnit 2

Refrigerant Piping Materials

Pipe Size (inch)	Total Length (feet)	Number of Bends
1/4	30	2
3/8	30	2

Written Specifications

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General

QUALITY ASSURANCE

The units shall be listed by Electrical Testing Laboratories (ETL) and bear the ETL label.

All wiring shall be in accordance with the National Electrical Code (N.E.C.).

The units shall be manufactured in a facility registered to ISO 9001 and ISO14001 which is a set of standards applying to environmental protection set by the International Standard Organization (ISO).

All units must meet or exceed the 2010 Federal minimum efficiency requirements and the ASHRAE 90.1 efficiency requirements for VRF systems. Efficiency shall be published in accordance with the Air-Conditioning, Heating, and Refrigeration Institute (AHRI) Standard 1230.

System start-up supervision shall be a required service to be completed by the manufacturer or a duly authorized, competent representative that has been factory trained in system configuration and operation. The representative shall provide proof of manufacturer certification indicating successful completion within no more than two (2) years prior to system installation. This certification shall be included as part of the equipment and/or controls submittals.

DELIVERY, STORAGE AND HANDLING

Unit shall be stored and handled according to the manufacturer's recommendation.

Warranty

The CITY MULTI units shall be covered by the manufacturer's limited warranty for a period of one (1) year parts and seven (7) year compressor to the original owner from date of installation.

Installing contractor shall meet manufacturer requirements to obtain extended manufacturer's limited parts and compressor warranty for a period of ten (10) years to the original owner from date of installation. This warranty shall not include labor.

Manufacturer shall have a minimum of fifteen (15) years continuous experience providing VRF systems in the U.S. market.

All manufacturer technical and service manuals must be readily available for download by any local contractor should emergency service be required. Registering and sign-in requirements which may delay emergency service reference are not allowed.

The CITY MULTI VRF system shall be installed by a contractor with extensive CITY MULTI install and service training. The mandatory contractor service and install training should be performed by the manufacturer.

Controls

OVERVIEW

The control system shall consist of a low voltage communication network and a web-based interface. The controls system shall gather data and generate web pages accessible through a conventional web browser on each PC connected to the network. Operators shall be able to perform all normal operator functions through the web browser interface.

Furnish energy conservation features such as optimal start, request-based logic, and demand level adjustment of overall system capacity as specified in the sequence.

System shall be capable of email generation for remote alarm annunciation.

ELECTRICAL CHARACTERISTICS

General:

Controller power and communications shall be via a common non-polar communications bus and shall operate at 30VDC.

Wiring:

Control wiring shall be installed in a daisy chain configuration from indoor unit to indoor unit, to the BC controller (main and subs, if applicable) and to the outdoor unit. Control wiring to remote controllers shall be run from the indoor unit terminal block to the controller associated with that unit.

Control wiring for centralized controllers shall be installed in a daisy chain configuration from outdoor unit to outdoor unit, to the system controllers (centralized controllers and/or integrated web based interface), to the power supply.

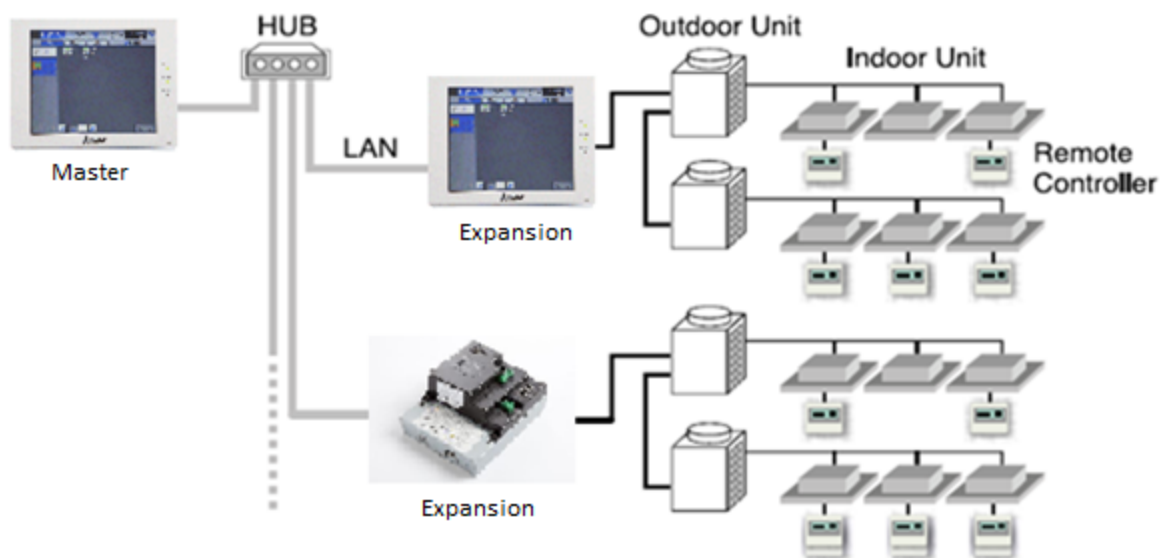
Wiring type:

Wiring shall be 2-conductor (16 AWG), twisted, stranded, shielded wire as defined by the Diamond System Builder output.

Network wiring shall be CAT-5 with RJ-45 connection.

CITY MULTI CONTROLS NETWORK

The CITY MULTI Controls Network (CMCN) consists of remote controllers, centralized controllers, and/or integrated web based interface communicating over a high-speed communication bus. The CITY MULTI Controls Network shall support operation monitoring, scheduling, occupancy, error email distribution, personal web browsers, tenant billing, online maintenance support, and integration with Building Management Systems (BMS) using either LonWorks® or BACnet® interfaces. The below figure illustrates a sample CMCN System Configuration.



CMCN System Configuration

CMCN: REMOTE CONTROLLERS

Wireless MA Remote Controller

The Wireless MA Remote Controller used in conjunction with the Wireless MA Remote Controller Receiver shall be capable of controlling up to 16 indoor units (defined as 1 group).

The Wireless MA Remote Controller and Receiver shall only be used in the same group with other Wireless MA or Simple MA Remote Controllers, with up to two remote controllers per group.

Wireless MA Remote Controller			
Item	Description	Operation	Display
ON/OFF	Run and stop operation for a single group	Each Group	Each Group
Operation Mode	Switches between Cool/Dry/Auto/Fan/Heat. Operation modes vary depending on the air conditioner unit. Auto mode is in the R2/WR2-Series only.	Each Group	Each Group
Temperature Setting	Sets the temperature from 57°F – 87°F depending on operation mode and indoor unit.	Each Group	Each Group

Fan Speed Setting	Available fan speed settings depending on indoor unit.	Each Group	Each Group
Air Flow Direction Setting	Air flow direction angles (4 or 5 angle Swing) Auto Louver ON/OFF Air flow direction settings vary depending on the indoor unit model.	Each Group	Each Group
Timer Operation	One ON/OFF setting can be set for one day.	Each Group	Each Group
Permit / Prohibit Local Operation	Individually prohibit operation of each local remote control function (ON/OFF, Change operation mode, Set temperature, Reset filter). *1. If operation is performed when the local remote controller inactivation command is received from the main system controller, a buzzer will ring and an LED will flash.	N/A	Each Group *1
Display Indoor Unit Intake Temp	Measures the intake temperature of the indoor unit when the indoor unit is operating.	N/A	N/A
Error	When an error is currently occurring on an air conditioner unit, the operation lamp on the signal receiving unit will flash.	N/A	Each Group
Test Run	Operates air conditioner units in test run mode.	Each Group	Each Group
Ventilation Equipment	Up to 16 indoor units can be connected to an interlocked system that has one LOSSNAY unit.	N/A	N/A

Equipment Schedules

MITSUBISHI ELECTRIC TRANE HVAC US: CITY MULTI VRF OUTDOOR UNIT SCHEDULE

System Tag	Tag Reference	M-NET Address	Model Number	Modules	Nominal Cooling Capacity (BTU/h)	Nominal Heating Capacity (BTU/h)	Design Cooling Outdoor Temp DB (°F)	Design Heating Outdoor Temp WB (°F)	Corrected Cooling Total Capacity (BTU/h)	Corrected Heating Capacity (BTU/h)	Electrical-Per Module				Notes / Options
											208/230 or [460V]				
											Voltage / Phase	MCA 208/230 or [460V]	RFS	MOCP	
System 1	OutdoorUnit 1	N/A	MXZ-2C20NA4-U1		18,000	22,000	95.0	43.0	17,838.7	20,986.5	208/230V / 1-phase	17.2	20	20	1, 2, 3, 4, 5

Notes & Options:

- 1 Nominal cooling capacities are based on indoor coil EAT of 80/67°F (DB/WB), outdoor of 95°F (DB)
- 2 Nominal heating capacities are based on indoor coil EAT of 70°F (DB), outdoor of 43°F (WB)
- 3 Efficiency values for EER, IEER, COP are based on AHRI 1230 test method for mixture of ducted & non-ducted indoor units.
- 4 For systems with multiple modules, refrigerant pipe dimensions indicate total system combined piping downstream of module twinning.
- 5 Added field charge listed is in addition to factory charge, this must be updated based upon final as-built piping layout.

mitsubishi electric trane hvac us: city multi vrf indoor unit schedule

System Tag	Room Name	Tag Reference	Model	Type	Nominal Cooling Capacity (BTU/h)	Nominal Heating Capacity (BTU/h)	Cooling Design Entering Temp DB/WB (°F) / [Water in temp]	Heating Design Entering Temp DB/WB (°F) / [Water in temp]	Corrected Capacity					Refrig Pipe Dim Liquid/Suction (inch)	Fan Speed Setting	Peak Fan Airflow (cfm) / [Design gpm G(US)/min]	Max Fan ESP Setting 208V/230V (IN WG)	Voltage / Phase	Electrical MCA/MFS	Notes / Options
									Cooling Diversity Full/Partial (See Note 5, 6)	Cooling Total Capacity (BTU/h)	Cooling Sensible Capacity (BTU/h)	Heating Diversity Full/Partial (See Note 5, 6)	Heating Capacity (BTU/h)							
System 1	front room 2	IndoorUnit 1	MFZ-KJ09NA-U1	Floor-Standing Type (Exposed)	9,000	10,500	80.0/67.0	70	FULL DEMAND	8,945.0	7,096.9	FULL DEMAND	10,500.0	1/4 / 3/8	HIGH	417		208/230V/1-phase	Powered by Outdoor	1, 2, 3, 4
System 1	Bedroom	IndoorUnit 2	MFZ-KJ09NA-U1	Floor-Standing Type (Exposed)	9,000	10,500	80.0/67.0	70	FULL DEMAND	8,893.7	7,077.1	FULL DEMAND	10,486.5	1/4 / 3/8	HIGH	417		208/230V/1-phase	Powered by Outdoor	1, 2, 3, 4

Notes & Options:

- 1 Nominal cooling capacities are based on indoor coil EAT of 80/67°F (DB/WB), outdoor of 95°F (DB)
- 2 Nominal heating capacities are based on indoor coil EAT of 70°F (DB), outdoor of 43°F (WB)
- 3 See outdoor unit schedule for outdoor ambient conditions, connected capacity, and other factors associated with corrected capacities
- 4 See schematic piping/control diagram for indication of required indoor unit remote controllers, system controllers, and integration devices.
- 5 Full demand corrected capacity includes de-rate associated with indoor vs. outdoor connected capacity indicated on outdoor unit schedule for associated system.
Partial corrected capacity assumes sufficient diversity exists such that the connected capacity de-rate does not apply.
It is the designer's responsibility to ensure "Diamond System Builder" is set in the appropriate output capacity setting (full demand/partial demand) prior to generating this schedule.
- 6 It is recommended to always base heating corrected capacity on full demand.

Design View Piping Diagrams

Indoor Units: 2 / 2 to 2
 Capacity: 18 / 10 to 24
 * Connectable capacity is not actual capacity.
 Total Pipe Length: 32.0 / 164.0 feet
 Furthest Actual: 30.0 / 82.0 feet
 Furthest Equiv.: 32.0 / 82.0 feet

Correction Factors

Outdoor Unit Capacity: 1.00 1.00
 Temperature: 0.99 1.00
 Piping Length: 0.99 1.00
 Defrosting: - 1.00
 User Derate: 1.00 1.00

Total Derate: 0.99 1.00
 Additional Refrigerant: 0.00 lb
 Total Refrigerant Amount: 5.93 lb

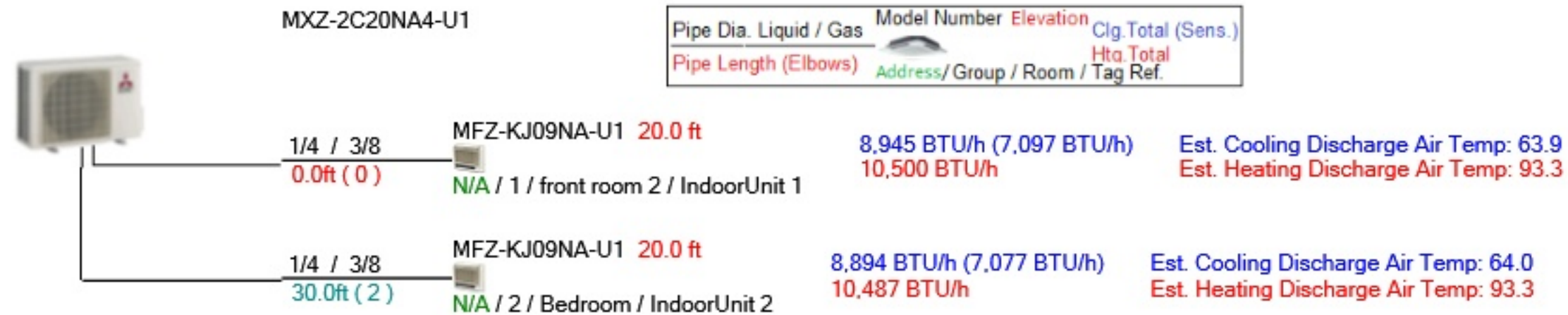
Conditions (°F)

Cooling

Indoor DB 80.0 Humidity 51.8% Indoor WB 67.0
 Outdoor DB 95.0

Heating

Indoor DB 70.0
 Outdoor DB 47.0 Humidity 72.8% Outdoor WB 43.0



AutoCAD Piping & Wiring Diagrams

Tunisha Riley

DIAGRAM SYMBOL LEGEND	
DISPLAY	DESCRIPTION
—//—	POWER WIRE
—	CONTROL WIRE
—	REF. PIPE

CONT.No		PAGE	
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CITY MULTI SYSTEM SCHEMATIC DWG.

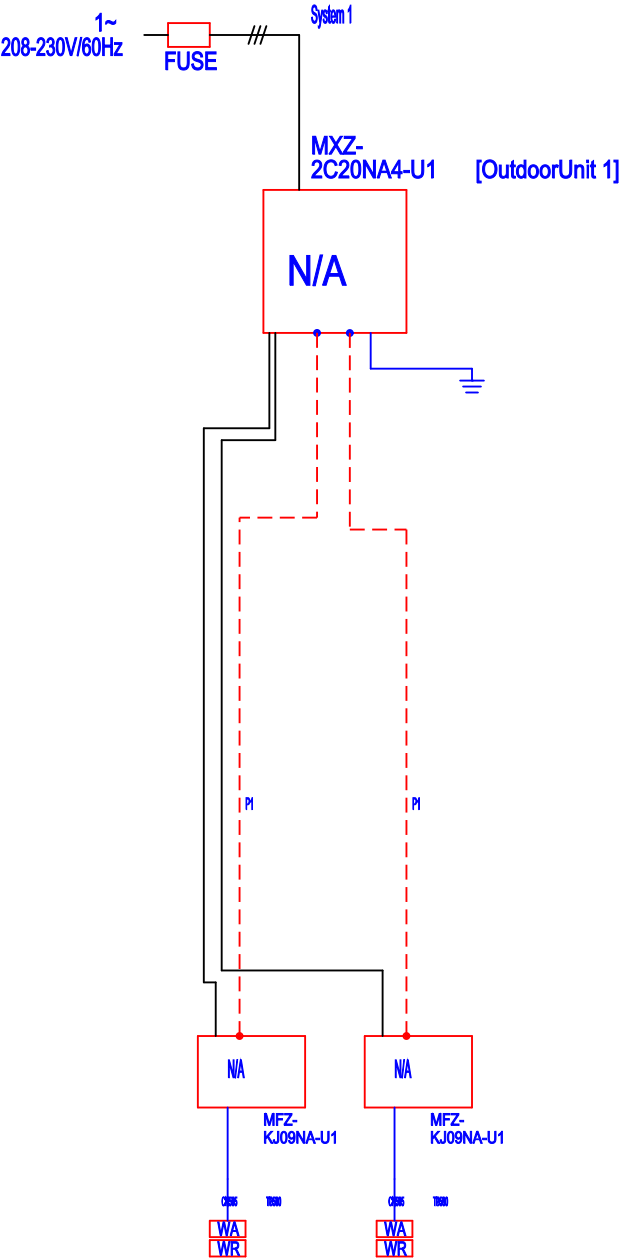
This drawing is schematic in nature. Final routing of piping & wiring shall be determined by the installing contractor and/or designer of record

Additional refrigerant charge is needed depending on the size and length of extended piping. Please refer the amount of pre-charge and the formula of calculation which is mentioned on the data book.

1.25mm²(16 AWG) : 1.25mm²(16 AWG) or more. 0.75mm²(20 AWG) : between 0.5mm²(24 AWG) and 0.75mm²(20 AWG). 2

Coded Notes:
NOTE 1: Install twinning Y's within 15 degrees of level and with 20 inches of straight pipe on converging connection - reference installation manual for additional details including but not limited to special trapping requirements when twinning, and pipe slope requirements

PIPING AND CONTROLS	
SYMBOL	LIQUID PIPE/GAS PIPE SIZE
P1	1/4 / 3/8
SYMBOL	MODEL NUMBER
WAWR	stock controller



front room 2 Bedroom
IndoorUnit 1 IndoorUnit 2

REMARKS
Originator: Derrell Black
Comments:

Submittal Documents

MXZ-2C20NA4
1.66-TON MULTI-ZONE INVERTER HEAT-PUMP SYSTEM

HEATING & AIR CONDITIONING

Job Name:

Date:

System Reference:



FEATURES

- Variable speed INVERTER-driven compressor
- M-NET connection through the outdoor unit
- Quiet outdoor unit operation as low as 50 dB(A)
- Intelligent Power Module (IPM) for reduced power loss
- Selectable thermal lockout/restart function
- Optional base pan heater
- High-pressure switch for additional protection
- A thermal differential of +/-1° F from setpoint
- Blue Fin anti-corrosion treatment applied to the outdoor unit heat exchanger for increased coil protection and longer life
- Rated for 2,000 hours spraying time per ASTM B117 Standard

ENERGY STAR products are third-party certified by an EPA-recognized Certification Body.

SPECIFICATIONS: MXZ-2C20NA4

Cooling ¹ (Non-Ducted // Mix (Mid-static) // Ducted (Mid-static) Mix (High-static) // Ducted (High-static))	Maximum Capacity	BTU/H	20,000 // 21,000 // 22,000 20,000 // 20,000
	Rated Capacity	BTU/H	18,000 // 19,000 // 20,000 19,000 // 20,000
	Minimum Capacity	BTU/H	5,700 // 5,700 // 5,700 6,550 // 7,400
	Maximum Power Input	W	2,245 // 2,258 // 2,270 2,258 // 2,270
	Rated Power Input	W	1,417 // 1,709 // 2,000 1,709 // 2,000
	Power Factor (208V, 230V)	%	99.0, 99.0 // 99.0, 99.0 // 99.0, 99.0 99.0, 99.0 // 99.0, 99.0
Heating at 47°F ² (Non-Ducted // Mix (Mid-static) // Ducted (Mid-static) Mix (High-static) // Ducted (High-static))	Maximum Capacity	BTU/H	25,500 // 25,000 // 24,500 25,000 // 24,500
	Rated Capacity	BTU/H	22,000 // 22,000 // 22,000 22,000 // 22,000
	Minimum Capacity	BTU/H	7,400 // 7,400 // 7,400 9,450 // 11,500
	Maximum Power Input	W	2,455 // 2,433 // 2,410 2,433 // 2,410
	Rated Power Input	W	1,641 // 1,706 // 1,771 1,706 // 1,771
	Power Factor (208V, 230V)	%	99.0, 99.0 // 99.0, 99.0 // 99.0, 99.0 99.0, 99.0 // 99.0, 99.0
Heating at 17°F ³ (Non-Ducted // Mix (Mid-static) // Ducted (Mid-static) Mix (High-static) // Ducted (High-static))	Maximum Capacity	BTU/H	15,500 // 15,000 // 14,500 15,000 // 14,500
	Rated Capacity	BTU/H	13,500 // 13,000 // 12,500 13,000 // 12,500
	Maximum Power Input	W	1,750 // 1,780 // 1,810 1,780 // 1,810
	Rated Power Input	W	1,364 // 1,515 // 1,665 1,515 // 1,665
	Maximum Capacity	BTU/H	11,100 // 11,000 // 10,900 11,000 // 10,900
	Maximum Power Input	W	1,600 // 1,650 // 1,700 1,650 // 1,700
Efficiency (Non-Ducted // Mix (Mid-static) // Ducted (Mid-static) Mix (High-static) // Ducted (High-static))	SEER2		20.0 // 18.0 // 16.0 18.0 // 16.0
	EER2 ¹		12.7 // 11.35 // 10.0 11.35 // 10.0
	HSPF2 (IV)		9.7 // 9.4 // 9.1 11.0 // 9.1
	COP at 47°F ²		3.92 // 3.78 // 3.64 3.78 // 3.64
	COP at 17°F at Maximum Capacity ³		2.6 // 2.48 // 2.35 2.48 // 2.35
	COP at 5°F at Maximum Capacity ⁴		2.22 // 2.05 // 1.88 2.05 // 1.88
Electrical	ENERGY STAR® Certified		Yes // No // No No // No
	Electrical Power Requirements	Voltage, Phase, Frequency	208/230, 1, 60
	Guaranteed Voltage Range	V AC	187-253
	Voltage: Indoor - Outdoor, S1-S2	V AC	208/230
	Voltage: Indoor - Outdoor, S2-S3	V DC	24
	Short-circuit Current Rating (SCCR)	kA	5
Outdoor unit	Recommended Fuse/Breaker Size	A	20
	Recommended Wire Size (Indoor - Outdoor)	AWG	14
	Minimum Circuit Ampacity	A	17.2
	Maximum Overcurrent Protection	A	20
	Fan Motor Full Load Amperage	A	1.77
	Airflow Rate (Cooling / Heating)	CFM	1,342 / 1,458
	Refrigerant Control		LEV
	Defrost Method		Reverse Cycle
	Heat Exchanger Type		Plate Fin Coil
	Heat Exchanger Coating		Blue Fin Coating
	Sound Pressure Level, Cooling ¹	dB(A)	50
	Sound Pressure Level, Heating ²	dB(A)	54
	Compressor Type		DC INVERTER-driven Twin Rotary
	Compressor Model		SNB140FQUH2T
	Compressor Rated Load Amps	A	10.7
	Compressor Locked Rotor Amps	A	15.5
	Compressor Oil Type // Charge	oz.	NEO22 // 20.3
	Base Pan Heater		Optional
	Unit Dimensions	W: In. [mm]	33-1/16 [840]
		D: In. [mm]	13 [330]
		H: In. [mm]	27-15/16 [710]
	Package Dimensions	W: In. [mm]	40 [1,006]
		D: In. [mm]	18-1/2 [466]
		H: In. [mm]	35-1/8 [892]
	Unit Weight	Lbs.[kg]	126 [57]
	Package Weight	Lbs.[kg]	150 [68]
Outdoor unit operating temperature range	Cooling Intake Air Temp (Maximum / Minimum ⁵)	°FDB	115 / 14
	Cooling Thermal Lock-out / Re-start Temperatures	°FDB	10.4 / 14
	Heating Intake Air Temp (Maximum / Minimum)	°FWB	65 / 5
	Heating Thermal Lock-out / Re-start Temperatures	°FDB	-4 / 5

NOTES:

AHRI Rated Conditions ¹Cooling (Indoor // Outdoor) °F 80 DB, 67 WB // 95 DB, 75 WB
(Rated data is determined at a fixed compressor speed) ²Heating at 47°F (Indoor // Outdoor) °F 70 DB, 60 WB // 47 DB, 43 WB
³Heating at 17°F (Indoor // Outdoor) °F 70 DB, 60 WB // 17 DB, 15 WB

Conditions ⁴Heating at 5°F (Indoor // Outdoor) °F 70 DB, 60 WB // 5 DB, 4 WB

*Applications should be restricted to comfort cooling only; equipment cooling applications are not recommended for low ambient temperature conditions.

⁵A 5°F DB - 115°F DB when optional wind baffles are installed

For actual capacity performance based on indoor unit type and number of indoor units connected, please refer to MXZ Operational Performance.

Although the maximum connectable capacity is 130%, the outdoor unit cannot provide more than 100% of the rated capacity. Please utilize this over capacity capability for load shedding or applications where it is known that all connected units will NOT be operating at the same time.

Low, mid and high external static pressure tests conducted at 0.1, 0.3 and 0.5 in.w.g. respectively, according to AHRI 210/240. The external static pressures utilized have no bearing on the capabilities of the indoor unit; please refer to the indoor unit manual to select the correct external static pressure setting for the application.

SPECIFICATIONS: MXZ-2C20NA4

Refrigerant	Type		R410A
	Pre-Charged Refrigerant Amount	Lbs, oz	5.0, 15.0
	Maximum Pre-Charged Piping Length	Ft. [m]	131.0 [40.0]
	Additional Refrigerant Charge Per Additional Piping Length	oz./Ft. [g/m]	0.216 [20]
Indoor unit connection	Maximum Number of Connected IDU		2
	Minimum Number of Connected IDU		2
	Minimum connected capacity	BTU/H	12,000
	Maximum connected capacity	BTU/H	24,000
Piping	Liquid Pipe Size O.D. (Flared)	In.[mm]	A,B: 1/4 [A,B: 6.35]
	Gas Pipe Size O.D. (Flared)	In.[mm]	A,B: 3/8 [A,B: 9.52]
	Total Piping Length	Ft. [m]	164 [50]
	Maximum Height Difference, ODU above IDU	Ft. [m]	33 [10]
	Maximum Height Difference, ODU below IDU	Ft. [m]	49 [15]
	Farthest Piping Length from ODU to IDU	Ft. [m]	82 [25]
	Maximum Number of Bends for IDU		50

NOTES:

AHRI Rated Conditions ¹Cooling (Indoor // Outdoor) °F 80 DB, 67 WB // 95 DB, 75 WB
(Rated data is determined at a fixed compressor speed) ²Heating at 47°F (Indoor // Outdoor) °F 70 DB, 60 WB // 47 DB, 43 WB
³Heating at 17°F (Indoor // Outdoor) °F 70 DB, 60 WB // 17 DB, 15 WB

Conditions ⁴Heating at 5°F (Indoor // Outdoor) °F 70 DB, 60 WB // 5 DB, 4 WB

^{*}Applications should be restricted to comfort cooling only; equipment cooling applications are not recommended for low ambient temperature conditions.

^{**}A 5°F DB - 115°F DB when optional wind baffles are installed

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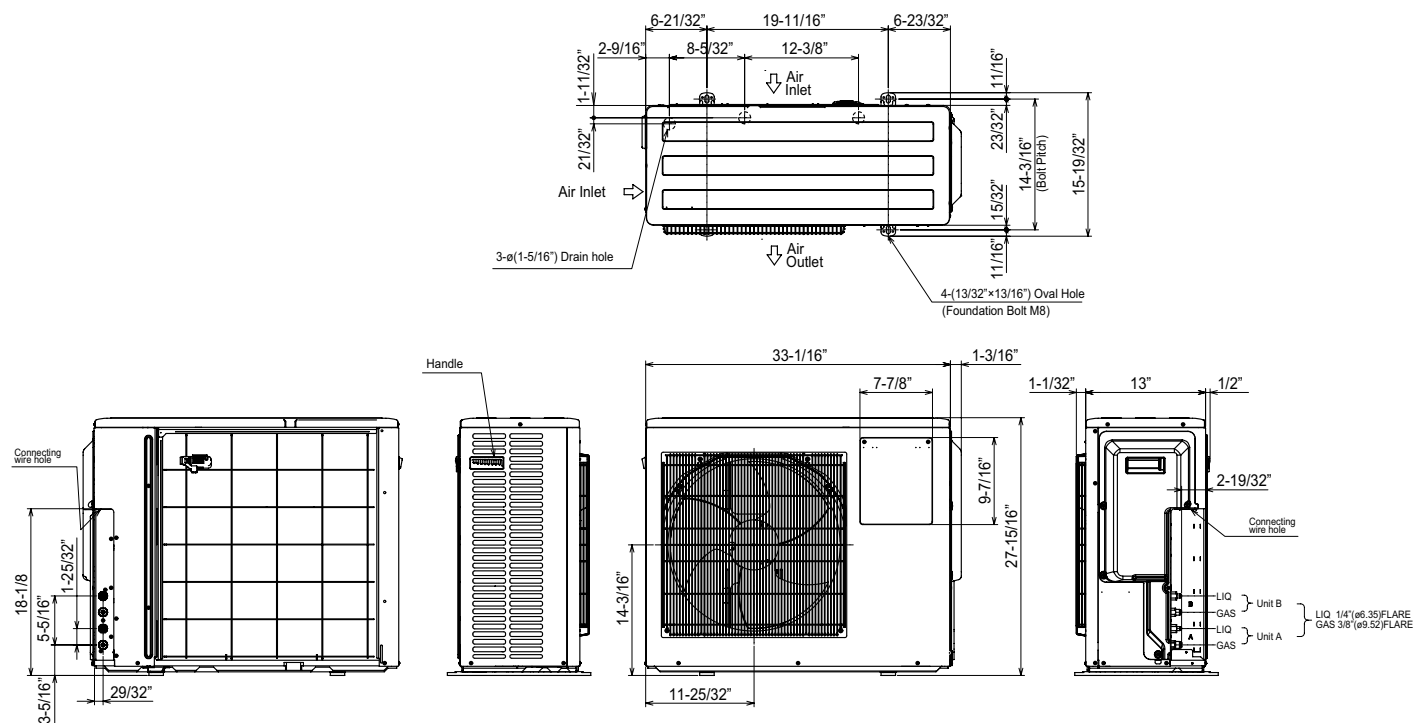
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OUTDOOR UNIT ACCESSORIES: MXZ-2C20NA4

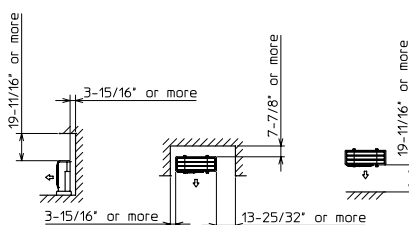
Air Outlet Guide	Air Outlet Guide	☐ MAC-856SG
Ball Valve	Refrigeration Ball Valve - 1/2"	☐ BV12FFSI2
	Refrigeration Ball Valve - 1/4"	☐ BV14FFSI2
	Refrigeration Ball Valve - 3/8"	☐ BV38FFSI2
	Refrigeration Ball Valve - 5/8"	☐ BV58FFSI2
Control Wire	M-Net Control Wire, 1,000' Roll (16-AWG, Standard, Twisted Pair, Shielded, Jacketed- Plenum rated)	☐ CW162S-1000
	M-Net Control Wire, 250' Roll (16-AWG, Standard, Twisted Pair, Shielded, Jacketed- Plenum rated)	☐ CW162S-250
Drain Socket	Drain Socket	☐ MAC-811DS
Hail Guards	Hail Guard	☐ HG-A8
M-NET Converter	M-NET Converter	☐ PAC-IF01MNT-E
Mini-Split Wire	14 Gauge, 4 wire MiniSplit Cable—250 ft. roll	☐ S144-250
	14 Gauge, 4 wire MiniSplit Cable—250 ft. roll	☐ SW144-250
	14 Gauge, 4 wire MiniSplit Cable—50 ft. roll	☐ S144-50
	14 Gauge, 4 wire MiniSplit Cable—50 ft. roll	☐ SW144-50
	16 Gauge, 4 wire MiniSplit Cable—250 ft. roll	☐ S164-250
	16 Gauge, 4 wire MiniSplit Cable—250 ft. roll	☐ SW164-250
	16 Gauge, 4 wire MiniSplit Cable—50 ft. roll	☐ S164-50
	16 Gauge, 4 wire MiniSplit Cable—50 ft. roll	☐ SW164-50
Mounting Pad	Condensing Unit Mounting Pad: 16" x 36" x 3"	☐ ULTRILITE1
	Outdoor Unit 3-1/4 inch Mounting Base (Pair) - Plastic	☐ DSD-400P
Optional Defrost Heater	Base Heater	☐ PAC-646BH-E
Port Adaptor	Adaptor: 1/2" x 3/8"	☐ MAC-A455JP-E
	Adaptor: 1/2" x 5/8"	☐ MAC-A456JP-E
	Adaptor: 3/8" x 1/2"	☐ MAC-A454JP-E
	Adaptor: 3/8" x 5/8"	☐ PAC-SG76RJ-E
Stand	18" Single Fan Stand	☐ QSMS1801M
	24" Single Fan Stand	☐ QSMS2401M
	Condenser Wall Bracket	☐ QSWB2000M-1
	Condenser Wall Bracket - Stainless Steel Finish	☐ QSWBSS
	Outdoor Unit Stand — 12" High	☐ QSMS1201M

OUTDOOR UNIT DIMENSIONS: MXZ-2C20NA4

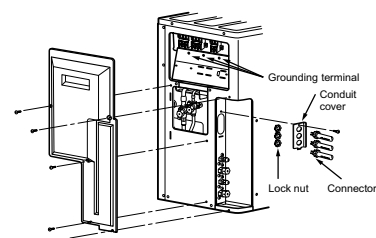
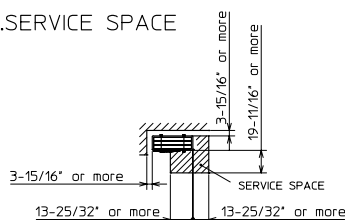
Unit: inch (mm)



1.FREE SPACE



2.SERVICE SPACE



1340 Satellite Boulevard Suwanee, GA 30024
Toll Free: 800-433-4822 www.mehvac.com

FORM# MXZ-2C20NA4 - 202311



MFZ-KJ09NA 9,000 BTU/H FLOOR-MOUNTED INDOOR UNIT



Job Name:

System Reference:

Date:



GENERAL FEATURES

- Floor-mounted indoor unit mounts on the floor or up to 5" above the floor
- Floor front panel access to the filter for ease of cleaning
- Perfect for difficult areas that may be smaller or don't have usable space on the walls
- Multiple fan speed options: Low, Medium, High, Powerful, Auto
- Quiet operation
- Multi-flow vane for faster heating
- Multiple control options available:
 - Hand-held Remote Controller (provided with unit)
 - kumo cloud® smart device app for remote access
 - Third-party interface options
 - Wired or wireless controllers
- Unit can be recessed mounted into wall

Specifications			System
Unit Type			MFZ-KJ09NA
Cooling Capacity ^{1,3}		BTU/H	9,000
Heating Capacity ^{2,3}		BTU/H	11,000
Electrical	Voltage, Phase, Frequency		208/230, 1, 60
	Guaranteed Voltage Range	V AC	187- 253V
	Voltage: Indoor - Outdoor, S1-S2	V AC	208/230
	Voltage: Indoor - Outdoor, S2-S3	V DC	24
	Short-circuit Current Rating [SCCR]	kA	5
Indoor Unit	MCA	A	1.0
	Fan Motor Full Load Amperage	A	0.62
	Airflow Rate at Cooling, Dry	CFM	275-219-208-173-138
	Airflow Rate at Cooling, Wet	CFM	234-213-177-147-117
	Airflow Rate at Heating, Dry	CFM	343-219-180-159-138
	Sound Pressure Level [Cooling]	dB[A]	38-34-30-25-21
	Sound Pressure Level [Heating]	dB[A]	41-32-27-24-21
	Drain Pipe Size	In. [mm]	5/8 O.D [15]
	Coating on Heat Exchanger		—
	External Finish Color		Munsell 1.0Y 9.2/0.2
	Unit Dimensions	W x D x H: In. [mm]	29-17/32 x 8-15/32 x 23-5/8 [750 x 215 x 600]
	Package Dimensions	W x D x H: In. [mm]	32-2/16 x 10-3/16 x 27-2/16 [816 x 275 x 693]
	Unit Weight	Lbs. [kg]	33 [15.0]
	Package Weight	Lbs. [kg]	41 [18.5]
Refrigerant	Type		R410A
Piping	Gas Pipe Size O.D. [Flared]	In. [mm]	3/8 [9.52]
	Liquid Pipe Size O.D. [Flared]	In. [mm]	1/4 [6.35]

NOTES:

Conditions

¹Cooling (Indoor // Outdoor)

°F 80 DB, 67 WB // 95 DB, 75 WB

²Heating at 47°F (Indoor // Outdoor)

°F 70 DB, 60 WB // 47 DB, 43 WB

³Capacity varies based on the number of indoor units operating and the model of the Multi-zone Outdoor Unit. For reference to connected capacity charts, please refer Multi-zone Outdoor Unit Operational Performance.:

INDOOR UNIT ACCESSORIES: MFZ-KJ09NA

Control Interface	BACnet® and Modbus® Interface	☐ PAC-UKPRC001-CN-1
	CN24 Relay Kit	☐ CN24RELAY-KIT-CM3
	IT Extender	☐ PAC-WHS01IE-E
	kumo station® for kumo cloud®	☐ PAC-WHS01HC-E
	Lockdown bracket for remote controller	☐ RCMKP1CB
	System Control Interface	☐ MAC-334IF-E
	Thermostat Interface	☐ PAC-US444CN-1
	Thermostat Interface	☐ PAC-US445CN-1
	USNAP Adapter	☐ PAC-WHS01UP-E
	Wireless Interface for kumo cloud®	☐ PAC-USWHS002-WF-2
Remote Sensor	Wired Remote Sensor	☐ M21EAA307
	Wireless temperature and humidity sensor for kumo cloud®	☐ PAC-USWHS003-TH-1
Wired Remote Controller	Deluxe Wired MA Remote Controller†	☐ PAR-40MAAU
	Simple MA Remote Controller†	☐ PAC-YT53CRAU-J
	Touch MA Controller†	☐ PAR-CT01MAU-SB
Wireless Remote Controller	kumo touch™ RedLINK™ Wireless Controller	☐ MHK2
Condensate	Blue Diamond (Advanced) Mini Condensate Pump w/ Reservoir & Sensor (208/230V) [recommended]	☐ X87-721
	Blue Diamond (MicroBlue) Mini Condensate Pump (110/208/230V) up to 18,000 BTU/H	☐ X86-003
	Blue Diamond Alarm Extension Cable — 6.5 Ft.	☐ C13-192
	Blue Diamond MultiTank — collection tank for use with multiple pumps	☐ C21-014
	Blue Diamond Sensor Extension Cable — 15 Ft.	☐ C13-103
	Refco Condensate Pump (100-240 VAC) up to 120,000 BTU/H	☐ COMBI
	Sauermann Condensate Pump	☐ SI30-230
Disconnect Switch	(30A/600V/UL) [fits 2" X 4" utility box] - Black	☐ TAZ-MS303
	(30A/600V/UL) [fits 2" X 4" utility box] - White	☐ TAZ-MS303W
Drain Hose	Flexible Mini-Split Drain Hose	☐ DRX-16
Filter	Anti-allergy Enzyme Filter	☐ MAC-408FT-E
Floor Mount Air Guide	Floor Mount Air Guide	☐ MAC-760FD-E
Lineset	15' x 1/4" x 15' / 3/8" Lineset (Twin-Tube Insulation)	☐ MLS143812T-15
	30' x 1/4" x 30' / 3/8" Lineset (Twin-Tube Insulation)	☐ MLS143812T-30
	50' x 1/4" x 50' / 3/8" Lineset (Twin-Tube Insulation)	☐ MLS143812T-50
	65' x 1/4" x 65' / 3/8" Lineset (Twin-Tube Insulation)	☐ MLS143812T-65

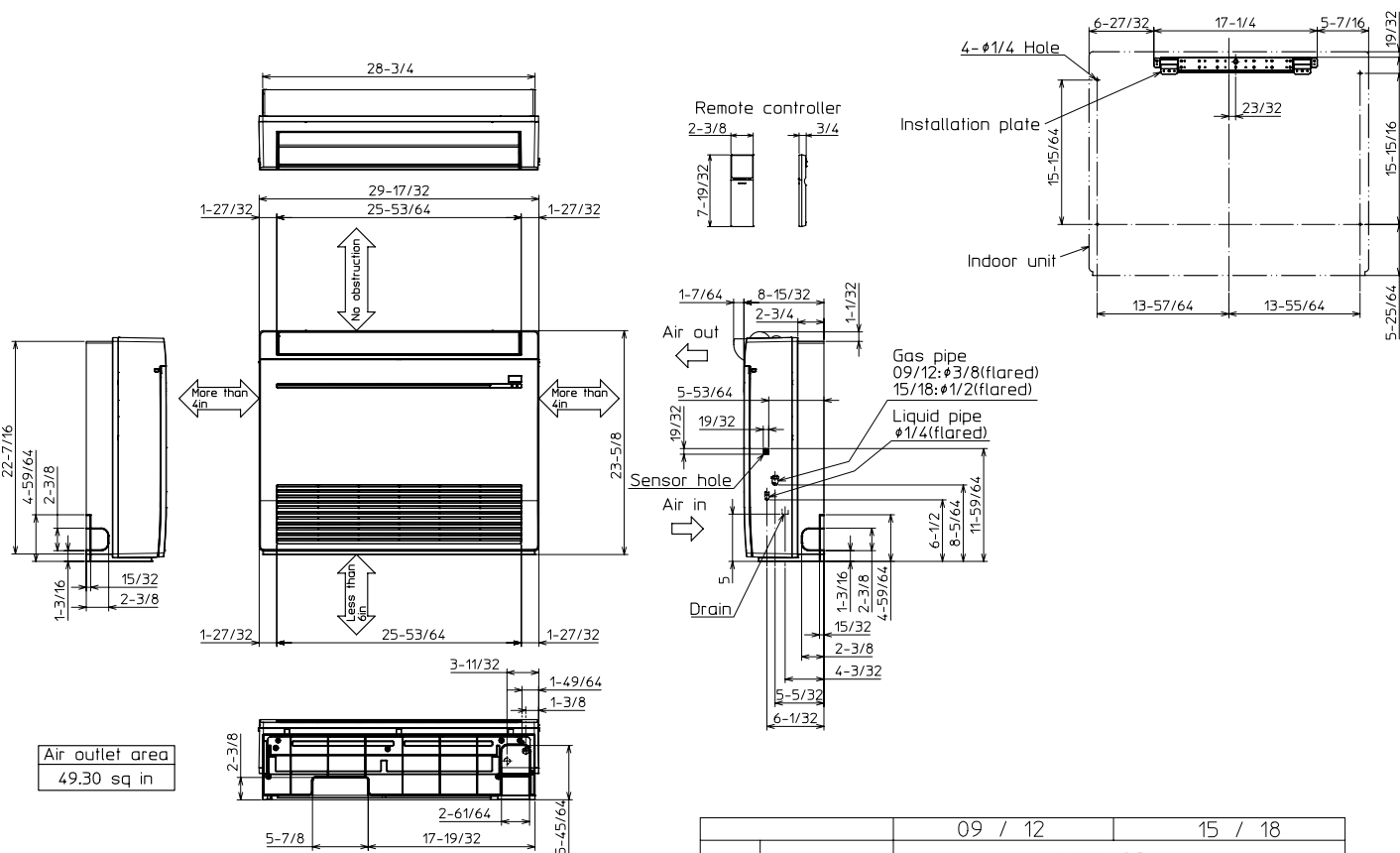
NOTES:

†Requires MAC-334IF-E

- M-Series EZ FIT® Recessed Ceiling Cassette, Floor-mount and Wall-mount
 - Allows indoor units to connect to an MA Controller:
 - Deluxe MA Remote Controller
 - Simple MA Controller
 - Touch MA Controller

INDOOR UNIT DIMENSIONS: MFZ-KJ09NA

Unit: inch



		09 / 12	15 / 18
Pipe cover	For liquid line	1-1/16 O.D	
	For gas line	1-1/16 O.D	1-7/32 O.D
Piping	Liquid line	Flared connection 1/4	
	Gas line	Flared connection 3/8	Flared connection 1/2
Drain hose		Heat insulator 1-9/64 O.D	Connection point 5/8 O.D Effective length 13-25/32 (case of right backward piping)

FORM# MFZ-KJ09NA-U1 - 202209

1340 Satellite Boulevard Suwanee, GA 30024
Toll Free: 800-433-4822 www.mehvac.com



Warranty Document

MITSUBISHI ELECTRIC TRANE HVAC US LLC

1340 Satellite Boulevard
Suwanee, GA 30024

Applicable Outdoor Models: SUZ-KA*NA2, SUZ-KA*NAR1, MUZ-JP*, MUFZ-KJ*, MUZ-GL*, MUY-GL*, MXZ-2~8C*, MUZ-FH*, MUZ-HE*, MUZ-D*, MUY-D*, MUZ-FS*, MUZ-GS*

Applicable Indoor Models: MSZ-JP*, MSZ-EF*, MSZ-GL*, MSY-GL*, MSZ-FH*, MSZ-HE*, SEZ-KD*NA4, SEZ-KD*NA4R1, MFZ-KJ*, SLZ-KA*NAR1, SLZ-KF*, SVZ-KP*, MVZ-A*, MLZ-KP*, MSZ-D*, MSY-D*, MSZ-GS*, MSZ-FS*

LIMITED WARRANTY STATEMENT

Mitsubishi Electric M-Series Split Air-conditioner and Heat-pump Systems

Subject to the terms and conditions of this Limited Warranty Statement (the "Limited Warranty"), MITSUBISHI ELECTRIC TRANE HVAC US LLC ("METUS") warrants to the original purchaser of this M-Series System (as used herein, "System" shall mean M-Series outdoor and indoor components connected via refrigerant piping and electrical wiring) purchased on or after **May 1, 2019** from a licensed HVAC contractor and installed by such contractor in the continental United States, Alaska and Hawaii, that:

- A. The parts are warranted for a period of five (5) years to the original owner of the System.** If any parts should prove defective due to improper workmanship and/or material for a period of five (5) years from the date of installation, METUS will replace any defective part without charge for the part. Replacement parts are warranted for the remainder of the original 5-year warranty period. Parts used for replacement may be of like kind and quality and may be new or remanufactured. Defective parts must be made available to METUS in exchange for the replacement parts and become the property of METUS.
- B. The compressor is warranted for a period of seven (7) years to the original owner of this System.** If the compressor should prove defective due to improper workmanship and/or material for a period of seven (7) years from the date of installation, METUS will replace the defective compressor without charge for the compressor. Replacement compressors are warranted for the remainder of the original 7-year warranty period. Compressors used for replacement may be of like kind and quality and may be new or remanufactured. Defective compressors must be made available to METUS in exchange for the replacement compressor and become the property of METUS.
- C. Notwithstanding the foregoing, if the System is installed in a residential single-family home and registered within 90 days from installation, the parts and compressor will be warranted for a period of 10 years, to the original owner, so long as the original owner resides in the home.** Specifically, if any parts and/or the compressor should prove defective due to improper workmanship and/or material for a period of ten (10) years from the date of installation, METUS will replace any defective parts or compressor without charge for the part or compressor. The replacement parts and/or compressor are warranted for the remainder of the original 10-year warranty period. Parts and/or compressors used for replacement may be of like kind and quality and may be new or remanufactured. Defective parts and/or compressors must be made available to METUS in exchange for the replacement parts and become the property of METUS.
- D. Notwithstanding the foregoing, if the System is installed in a residential single-family home by a DIAMOND CONTRACTOR and registered within 90 days from installation, the parts and compressor will be warranted for a period of 12 years, to the original owner, so long as the original owner resides in the home.** Specifically, if any parts and/or the compressor should prove defective due to improper workmanship and/or material for a period of twelve (12) years from the date of installation, METUS will replace any defective parts or compressor without charge for the part or compressor. The replacement parts and/or compressor are warranted for the remainder of the original 12-year warranty period. Parts and/or compressors used for replacement may be of like kind and quality and may be new or remanufactured. Defective parts and/or compressors must be made available to METUS in exchange for the replacement parts and become the property of METUS.
- E. NO LABOR.** These limited warranties do NOT include labor or any other costs incurred for service, maintenance, repair, removing, replacing, installing, complying with local building and electric codes, shipping or handling, or replacement of the System, compressors or any other parts. For items that are designed to be maintained or replaced by the original owner, the original owner is solely responsible for all labor and other costs of maintaining, installing, replacing, disconnecting or dismantling the System and parts (such as filters or belts) in connection with owner-required maintenance. Air filter cleaning and/or replacement for each applicable indoor unit are owner-required maintenance,

and labor for this procedure is not covered under warranty. Please consult the applicable technical documentation for air filter cleaning and other maintenance procedures.

- F. PROPER INSTALLATION.** This Limited Warranty applies only to Systems that are installed by contractors who are licensed for HVAC installation under applicable local and state law, and who install the Systems in accordance with (i) all applicable building codes and permits; (ii) METUS's installation and operation instructions; and (iii) good trade practices.
- G. REGISTRATION.** To receive certain benefits under this Limited Warranty, as set forth in more detail above, the product must be registered within 90 days of installation. To register, the original owner may complete and return the postage-paid registration card by U.S. mail or may register the product online at www.registermehvac.com. Diamond Contractors may submit product registration information online via the extranet portal. Residents of California or Quebec need not register the product to receive warranty benefits, but are encouraged to do so.

BEFORE REQUESTING SERVICE, please review the applicable technical documentation to insure proper installation and correct customer control adjustment for the System. If the problem persists, please arrange for warranty service.

1) TO OBTAIN WARRANTY SERVICE:

- a. Contact the licensed contractor who installed the System or the nearest licensed contractor, dealer, or distributor (whose name and address may be obtained on our website at www.mitsubishicomfort.com) of any defect within the applicable warranty time period.
- b. Proof of the installation date by a licensed contractor is required when requesting warranty service. Present the sales receipt, building permit or other document which establishes proof and date of installation. In the absence of acceptable proof, this Limited Warranty shall be deemed to begin one hundred twenty (120) days after the date of manufacture stamped on the System.
- c. This Limited Warranty applies only to Systems purchased on or after **May 1, 2019**, only while the System remains at the site of the original installation, and only to locations within the continental United States, Alaska and Hawaii.

- 2) THIS LIMITED WARRANTY DOES NOT COVER:** property damages, malfunction or failure of the System, or personal injury caused by or resulting from: (a) accident, abuse, negligence or misuse; (b) operating the System in a corrosive or wet environment, including those containing chlorine, fluorine or any other hazardous or harmful chemicals or environmental factors, including sea- or salt-water; (c) installation, alteration, repair or service by anyone other than a licensed contractor or other than pursuant to the manufacturer's instructions; (d) improper matching of System components; (e) improper sizing of the System; (f) improper or deferred maintenance contrary to the manufacturer's instructions; (g) physical abuse to or misuse of the System (including failure to perform any maintenance as described in the Operation manual such as air filter cleaning, or any System damaged by excessive physical or electrical stress; (h) Systems that have had a serial number or any part thereof altered, defaced or removed; (i) System used in any manner contrary to the Operation Manual; (j) freight damage; or (k) events of force majeure or damage caused by other external factors such as lightning, power surges, fluctuations in or interruptions of electrical power, rodents, vermin, insects, or other animal- or pest-related issues.

- 3) THIS LIMITED WARRANTY ALSO EXCLUDES:** (a) SERVICE CALLS WHERE NO DEFECT IN THE SYSTEM COVERED UNDER THIS WARRANTY IS FOUND; (b) System installation or set-ups; (c) Adjustments of user controls; (d) Systems purchased or installed outside the continental United States, Alaska and Hawaii; or (e) Systems purchased or installed prior to **May 1, 2019**. Consult the operating instructions for information regarding user controls.

- 4)** This Limited Warranty shall not be enlarged, extended or affected by, and no obligation or liability shall arise or grow out of, METUS providing, directly or indirectly, any technical advice, information and/or service to original owner in connection with the System.

- 5) EXCEPT AS OTHERWISE PROVIDED IN THIS LIMITED WARRANTY, METUS MAKES NO OTHER WARRANTIES OF ANY KIND WHATSOEVER REGARDING THE SYSTEM. METUS DISCLAIMS AND EXCLUDES ALL WARRANTIES NOT EXPRESSLY PROVIDED HEREIN AND ALL REMEDIES WHICH, BUT FOR THIS PROVISION, MIGHT ARISE BY IMPLICATION OR OPERATION OF LAW, INCLUDING, WITHOUT LIMITATION, THE IMPLIED WARRANTIES OF**

MERCHANTABILITY, NON-INFRINGEMENT OF THIRD PARTY RIGHTS, AND OF FITNESS FOR ANY PARTICULAR PURPOSE. NO ONE IS AUTHORIZED TO CHANGE THIS LIMITED WARRANTY IN ANY RESPECT OR TO CREATE ANY OTHER OBLIGATION OR LIABILITY FOR METUS IN CONNECTION WITH THE SYSTEM. METUS DISCLAIMS ALL LIABILITY FOR THE ACTS, OMISSIONS AND CONDUCT OF ALL THIRD PARTIES (INCLUDING, WITHOUT LIMITATION, THE INSTALLING CONTRACTOR) IN CONNECTION WITH OR RELATED TO THE SYSTEM.

- 6) UNDER NO CIRCUMSTANCES SHALL METUS BE LIABLE FOR ANY INDIRECT, INCIDENTAL, SPECIAL, PUNITIVE OR CONSEQUENTIAL DAMAGES INCLUDING, WITHOUT LIMITATION, INFRINGEMENT OF THIRD PARTY RIGHTS, LOST GOODWILL, LOST REVENUES OR PROFITS, WORK STOPPAGE, SYSTEM FAILURE, IMPAIRMENT OF OTHER GOODS, COSTS OF REMOVAL AND REINSTALLATION OF THE SYSTEM, LOSS OF USE, INJURY TO PERSONS OR PROPERTY ARISING OUT OR RELATED TO THE SYSTEM WHETHER BASED ON BREACH OF WARRANTY, BREACH OF CONTRACT, TORT OR OTHERWISE, EVEN IF METUS HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGE. IN NO EVENT SHALL METUS'S LIABILITY EXCEED THE ACTUAL PURCHASE PRICE OF THE SYSTEM WITH RESPECT TO WHICH ANY CLAIM IS MADE.**
- 7) SOME STATES DO NOT ALLOW LIMITATIONS ON WARRANTIES OR EXCLUSIONS OR LIMITATION OF DAMAGES, SO THE ABOVE LIMITATIONS OR EXCLUSIONS MAY NOT APPLY.**
- 8) DISPUTE RESOLUTION.** For any dispute with METUS, you agree to first contact us by phone (800-433-4822) or e-mail (CustomerCare@hvac.mea.com) or U.S. Mail at Mitsubishi Electric TRANE HVAC US LLC ATTN: Customer Care, 1340 Satellite Blvd., Suwanee, GA 30024, and attempt to resolve the dispute with us informally by providing your name, address, and contact information and describing the nature of the dispute. In the unlikely event that METUS has not been able to resolve a dispute with you within 60 days of your original informal claim (or sooner if, in METUS's opinion, a dispute is not likely to be resolved within 60 days), we each agree to resolve any claim, dispute, or controversy arising out of or in connection with or relating to this Limited Warranty, or the breach or alleged breach thereof (collectively, "Claims"), by binding arbitration before an arbitrator from Judicial Mediation and Arbitration Services ("JAMS") located in Gwinnett County, Georgia. JAMS may be contacted at www.jamsadr.com and will require you to pay an initial filing fee set by JAMS (unless you successfully apply for a waiver of this fee from JAMS). All other JAMS costs associated with the arbitration will be borne by METUS. The arbitration will be conducted in Gwinnett County, Georgia, unless you request an in-person hearing where you live, or if you and METUS agree otherwise. If the arbitrator decides in your favor, the award may include your costs of arbitration, your reasonable attorneys' fees and your reasonable costs for any expert and other witnesses, and any judgment on the award rendered by the arbitrator may be entered in any court of competent jurisdiction. If the arbitrator makes an award in your favor greater than METUS's last written offer, METUS will pay you the greater of the award or \$500, plus your reasonable attorney's fees, if any, and reimburse any reasonable expenses (including reasonable expert witness fees and costs) that are reasonably accrued for investigating, preparing, and pursuing your claim in arbitration, as determined by the arbitrator or as agreed to by you and METUS. Any judgment on the award rendered by the arbitrator may be entered in any court of competent jurisdiction. You may sue under state law in a small claims court of competent jurisdiction without first engaging in arbitration, but you must engage in arbitration before suing under the Federal Magnuson-Moss Act.
- 9) All claims must be brought in the parties' individual capacity, and not as a plaintiff or class member in any purported class or representative proceeding. This waiver applies to class arbitration unless such arbitration is necessary to effectuate the enforcement of the court class action waiver or in the event that class arbitration is expressly agreed to by METUS. You agree that you and METUS are each waiving the right to a trial by jury or to participate in a class action.**
- 10) You may opt-out of the foregoing arbitration and class action/jury trial waiver provision of this Limited Warranty by notifying METUS in writing within 30 days of purchase. Such written notification must be sent to Mitsubishi Electric TRANE HVAC US LLC ATTN: Legal, 5900-A Katella Avenue, Cypress, CA 90630, and must include (1) your name, (2) your address, (3) your warranted product's serial number, and (4) a clear statement indicating that you do not wish to resolve disputes through arbitration and demonstrating compliance with the 30-day time limit to opt-out.**
- 11) If any clause herein is found to be illegal or unenforceable, that clause will be severed from this Limited Warranty and the remainder of the Limited Warranty will be given full force and effect. As noted above, if a class action waiver of both court and arbitration class actions is found unenforceable, class arbitration will be expressly allowed under the Limited Warranty.**

- 12) Residents of California and Quebec do not need to register the product in order to get all of the rights and remedies of registered original owners under this warranty.
- 13) This Limited Warranty gives the original owner specific legal rights and the original owner may also have other rights that vary from state to state.
- 14) This Limited Warranty is valid only in the continental United States, Alaska and Hawaii, and it is not transferable.